O&M REPORT

Web-based tool for photovoltaic maintenance reporting

PROBLEM:

O&M REPORTING PROCESSES ARE MANUAL AND INCONSISTENT

WHAT IS STILL HAPPENING

- O&M reporting is still **manual and inconsistent**
- Multiple templates across stakeholders - **hard to standardize and compare**
- Data is **scattered** (Excel, Word/PDF, photos) - no single source of truth
- Missing structure - **incomplete or unreliable data**
- Delayed/manual uploads - **higher risk of errors and rework**

DAY-TO-DAY IMPACT

- Field technicians spend time on **reporting instead of operations**
- PMO / Project teams receive **inconsistent and delayed information**
- Slower validations, communication, and activity closure
- **Poor data quality** - impacts audits, performance analysis, and decision-making

GOAL

1

STANDARDIZE REPORTING

A unified structure with consistent sections and fields; reduces variability and ensures capture of essential information.

2

REDUCE MANUAL EFFORT

A guided web form replaces spreadsheets; users input only what is needed and reports are generated automatically.

3

IMPROVE DATA QUALITY

Defined fields reduce omissions; consistent naming conventions improve traceability and reliability for management.

SOLUTION:

REPORT HTML + CSS + JAVASCRIPT

1

HTML (STRUCTURE)

Defines fields and sections; e.g., dates, location, description, type (Preventive/Corrective/Cleaning).

2

CSS (LAYOUT)

Organizes layout for readability through alignment and spacing; e.g., separated sections and highlighted titles.

3

JAVASCRIPT

Automates behavior and outputs; e.g., show/hide fields, capture values, generate PDF, export to Excel.

PRODUCT

O&M TECHNICIANS

Needs: Quickly record interventions without ambiguity, ensuring all relevant fields are completed.

Value: Saves time in the field and reduces errors in documentation and scattered records.

Fewer Errors

PROJECT MANAGERS / PMO

Needs: Consistent reports to track execution, timelines, and stakeholder communication.

Value: Standardized deliverables (PDF) and structured data (Excel) to consolidate status and reduce follow-ups due to missing information.

Consistent Reports

ASSET MANAGERS

Needs: Reliable and comparable historical data across installations to support decision-making and audits.
Value: Data consistency, easy archiving, and structured exports for analysis and recurring reporting.

Reliable Historic Data

ENERGY COMPANIES

Needs: Replicable processes across sites and teams while maintaining reporting quality.

Value: Standardization and faster report generation, providing a foundation for future improvements (KPIs, storage, scaling).

Replicable Processes

LIVE DEMO

GitHub Pages:

https://anasalsas.github.io/O-M-Report-Generator-HTML-JavaScript-/

FUTURE IMPROVEMENTS

DATA VALIDATION

Add rules to ensure essential fields are completed and formats are correct (e.g., dates, required fields, consistent option sets). This improves dataset quality, reduces rework, and increases confidence in reporting.

Add or update content as required

KPIS DASHBOARD

Create a metrics view to track intervention volumes, types, trends, and operational indicators. This gives PMO and asset management quick visibility without needing to open individual reports. It complements exports by providing aggregated insights.

Add or update content as required